Integrating Renewable Energy into the U.S. Power Grid

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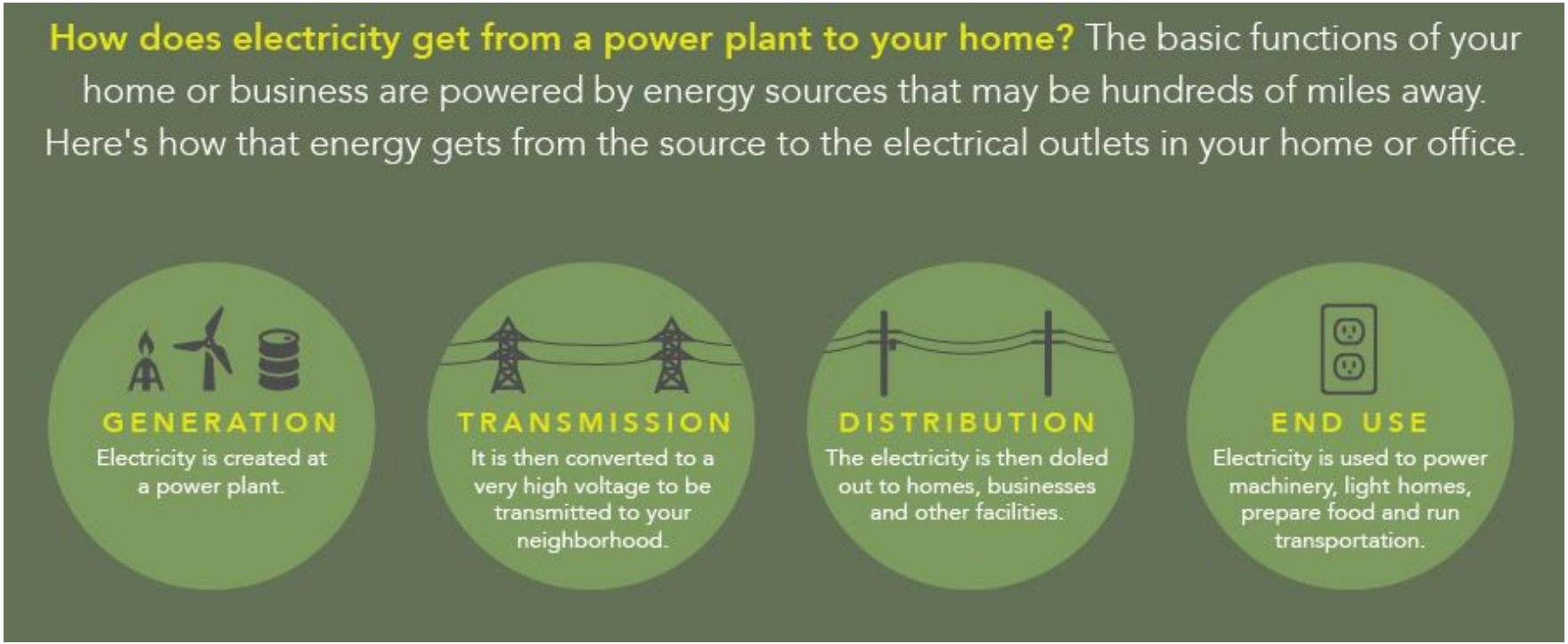
Terms and Definitions

- **Renewable energy sources** are defined as a source that produces little or no greenhouse emission and is environmentally sustainable.
- **Solar Power** or **Solar** is the conversion of energy from sunlight to electricity.
- **Wind Power** is the use of air currents to generate electricity.
- **AC** refers to Alternating Current.
- **DC** refers to Direct Current.



Source: Department of Energy

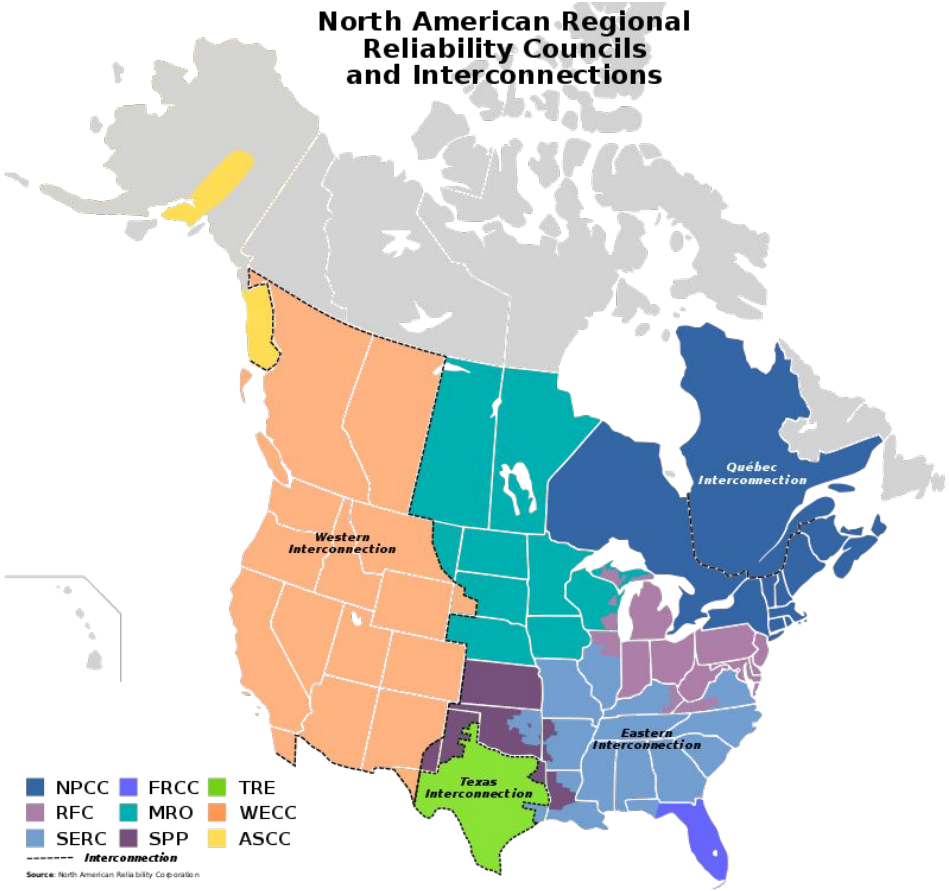
The U.S. Power Grid

Three main continental power grids.

More than 7,300 power plants.

160,000 miles of high-voltage power lines.

Millions of low-voltage power lines and distribution transformers.



An Aging System

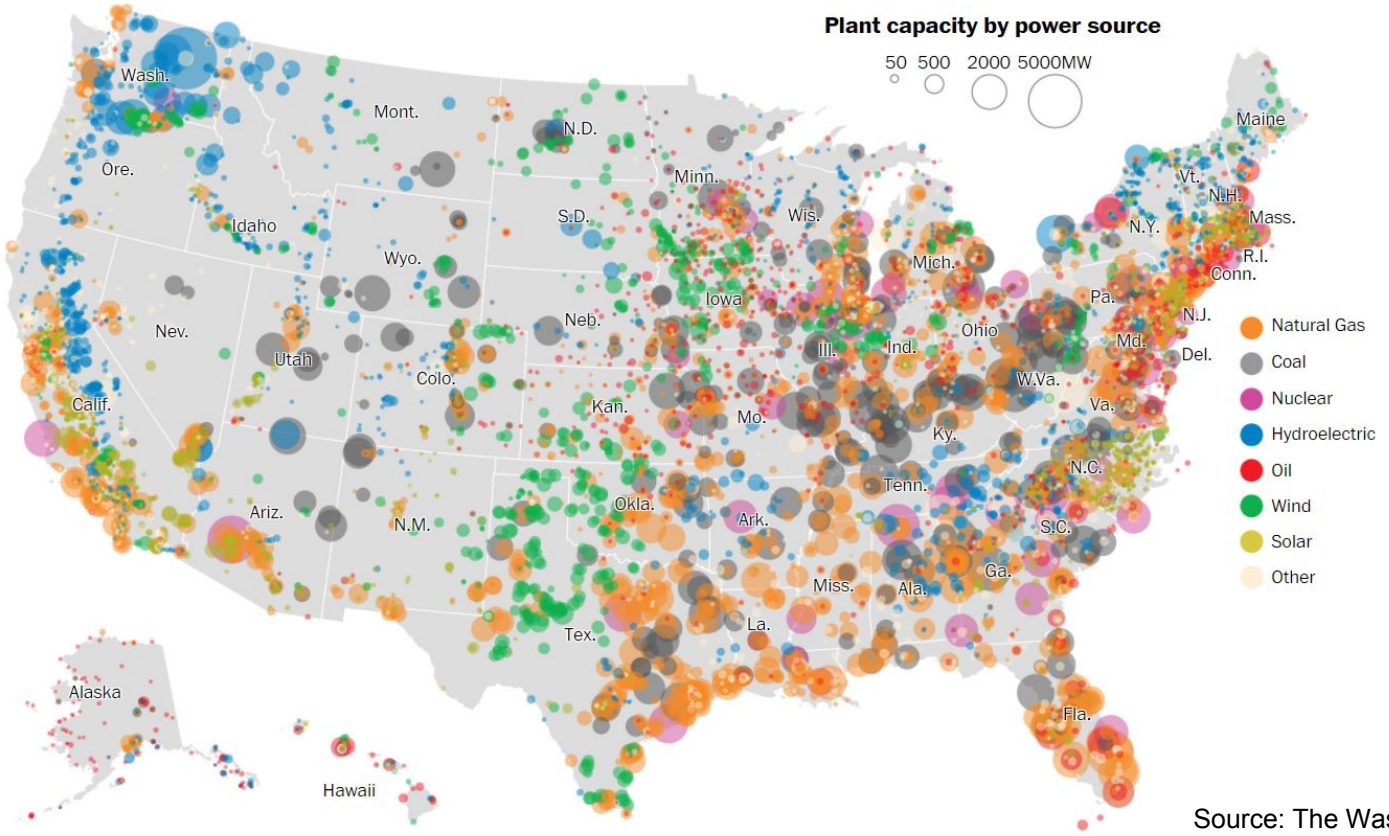
Most large infrastructure is from the **mid-20th century**, nearing end of life cycle.

Power plants are at their **maximum output** to meet a growing demand.

American Society of Civil Engineers rated the U.S. Power Grid in 2017 with a **D+**.

Estimated cost of modernization: **$5 trillion**

Traditional Energy Sources



Source: The Washington Post

Nonrenewable Energy Sources

34%

Natural Gas

Natural gas is formed when decomposing organic matter is subject to geological pressure. Harvested by fracking, which poses several risks.

- Release of toxic chemicals into the environment.
- Risk of earthquakes.
- Methane leaks.
- Uses a great deal of water.

30%

Coal

Coal is utilized on the east coast (south of NY) and in the midwest. As one of the oldest sources of energy, it is well integrated.

- Releases many greenhouse gases.
- Mercury emissions.
- Coal ash must be stored in special storage locations.
- Acid rain.

20%

Nuclear

While, nuclear power plants produce no greenhouse emissions, they generate a great deal of waste.

- Radioactive waste buried deep underground.
- Water cooled cores lead to water pollution.

1%

Oil

As a fossil fuel, oil is neither environmentally or economically sustainable.

- Fossil fuel, highly nonrenewable.
- Cost for industry goes up about 11% annually.

Renewable Energy Sources

Wind

Pros:

- Give economical value to rural areas
- Stable pricing
- Variety of designs and sizes

Cons:

- Kill 140,000 - 328,000 birds each year
- Mostly birds of prey

Solar

Pros:

- Easy to set up - connect to grid
- Integration into building materials
- Technology rapidly developing

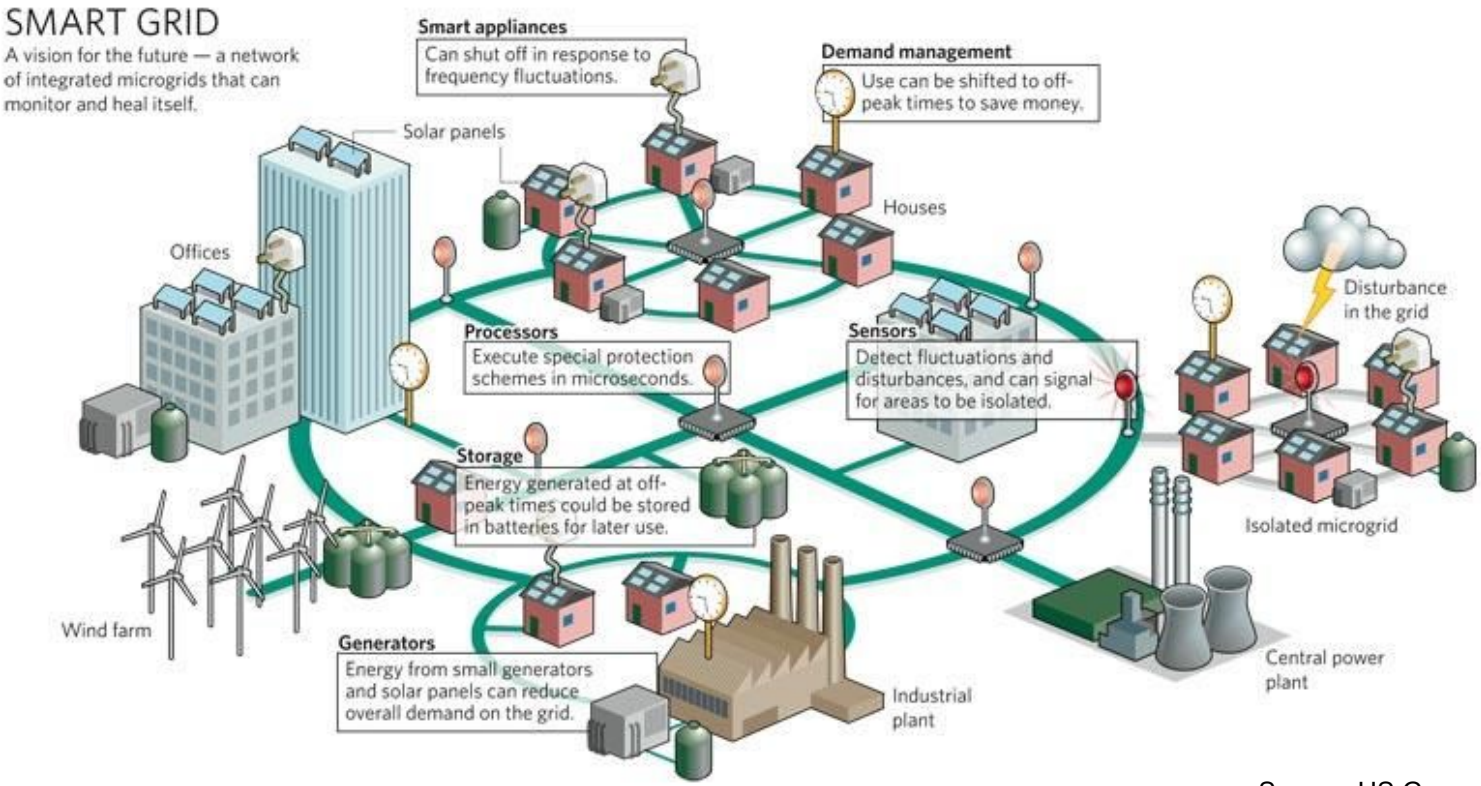
Cons:

- Harmful byproducts
- Transportation costs for byproducts

Technological Hurdles

- Long distance transmission
  - Renewables must be converted from DC to AC for long-distance transmission then back to DC
  - Conversions and transmission not efficient - renewables create more energy loss
- Lack of effective energy storage
  - Renewables are inconsistent - backed by energy storage
  - Less efficient than coal, natural gas, or nuclear plants
- Reliance on fossil fuels
  - Difficult to change current energy consumption habits
- Adaption of U.S. Smart Grid
  - Adapts to variability of demands throughout the day
  - Allows for offsets in energy across the region

Adapting the U.S. Smart Grid



Source: US Green Technology

Political Resistance

- Federal Regulations
- Approval from the interconnections' governing bodies
- Political lobbies

Source: CNBC

Economical Resistance

- Renewables are still more expensive
  - Need to be cheaper to be a proper replacement on a large scale
  - Prices are getting cheaper with technological advances
- Net metering is not favored by all utilities
  - 41 states have net metering rules implemented
  - Net metering rules are not consistent across states
- Fossil fuels are essential to the current economy
  - Cannot service today’s energy demand

Proposed Solution

- Integrating renewables into the grid
  - Overcoming cost differences
  - Technological advances
  - Implementing proper manufacturing policies
- Cuomo’s Reforming the Energy Vision (REV) plan for 2030
  - Offers guidelines and goals for power industry in NYS, an ultimatum
- Changing how we think about energy
  - Encouraging solar installation with positive net metering policies
  - Changing our consumption habits

Thank you! Questions?